YOUR
Functional Health



Guide to

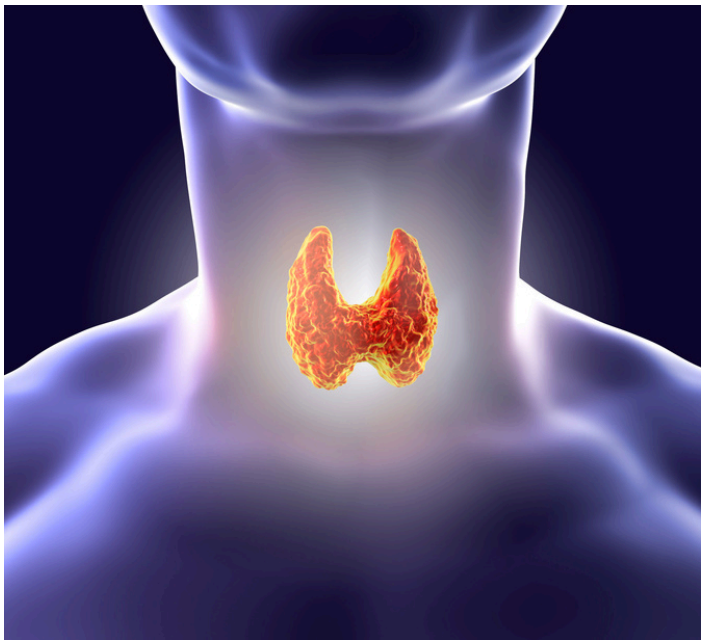
Thyroid Health

Keep your control center in good shape with this diet and lifestyle blueprint

THYROID 101

What you need to know

Your thyroid is a butterfly shaped gland located at the front of your neck. It is responsible for producing hormones (mainly T3 and T4) that work as messengers throughout the body to control your energy production, body temperature, muscle function, heart rate, and digestive function. You can think of the thyroid as the bodies thermostat and control system. If this organ is not working optimally, many symptoms arise.



Symptoms of Thyroid Trouble

Under-active Thyroid (Hypothyroidism)

- Fatigue
- Increased sensitivity to cold
- Constipation
- Dry skin
- Weight gain
- Poor bone density
- Hoarseness
- Muscle weakness
- Elevated blood cholesterol
- Muscle aches
- Painful joints
- Heavier or irregular menstrual periods
- Thinning hair
- Slowed heart rate
- Depression
- Impaired memory
- Enlarged thyroid gland (goiter)
- Fertility difficulties

Over-active Thyroid (Hyperthyroidism)

- Unintentional weight loss
- Fast heart rate
- Irregular heartbeat (arrhythmia)
- Palpitations
- Increased appetite
- Nervousness, anxiety and irritability
- Tremors
- Sweating
- Changes in menstrual patterns
- Increased sensitivity to heat
- Diarrhea
- Enlarged thyroid gland
- Fatigue
- Muscle weakness
- Difficulty sleeping
- Skin thinning
- Fine, brittle hair

TEAM THYROID

Whole Body System

The thyroid and the hormones it creates affects almost every other organ in the body. See how some other organ systems are altered by the health of the thyroid.



Heart

Power over heart rate, contractibility of the heart muscle, and even an affect on cholesterol levels.



Bone Health

Impact on the rate of bone replacement. Can lead to poor bone density.



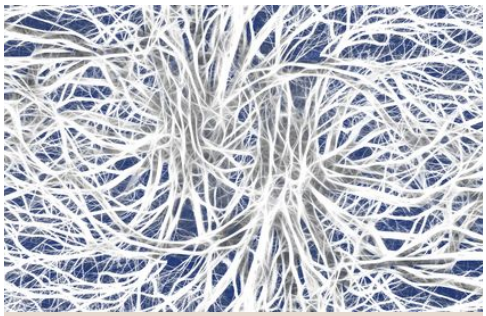
Muscle

Affects muscle repair and regeneration. Can lead to impaired skeletal muscle strength and muscle pain.



Digestive System

Impact on the rate in which the stomach empties and food is moved through the digestive tract. Vise Versa, poor gut health can have an impact on thyroid health.



Nervous System

Can change the rate and ability for signals and messages to be sent via the nervous system.



Reproductive System

Affects on production of various other hormones and has an impact on metabolic rate which can lead to changes in menstrual patterns and problems such as miscarriages.

TESTING FOR THYROID ISSUES

What you need to know



Many doctors will still only order TSH (thyroid stimulating hormone) when testing for thyroid issues. This is not a good indicator of whether or not there is a problem with the thyroid. Rather the following tests should be looked at as a whole. A doctor who is experienced with hormone health will be looking at the ratios between the following tests.

What a Full Thyroid Panel Includes

Even the best doctors may not order all of these tests the first time around. However TSH, Free T3, Free T4, and thyroid anti-bodies should be tested.

- TSH
- Free T3
- Free T4
- T3
- Thyroid Peroxidase (TPO) Antibodies
- Thyroid Antithyroglobulin Antibody (TAA)
- Reverse T3
- Thyroxine-binding Globulin (TBG)
- Thyroid-stimulating Immunoglobulin (TSI)

POTENTIAL CAUSES OF THYROID MALFUNCTION

Get to the Root Cause



Environmental Toxins

We all have daily exposure to toxins. Things like pesticides, herbicides, and chemicals found in our household and skincare products have a major impact our thyroid. In addition mold and heavy metals can cause thyroid dysfunction.



Stress

Consistently high cortisol levels can block hormone production and cause issues with thyroid function. This can be from emotional stressors as well has physical stress such as very high intensity exercise.



Nutrient Deficiency

Our thyroid needs the right building blocks to function. These nutrients are particularly important.

Vit D	**Selenium**
Vit D	**Iodine**
Omega 3	**Zinc**



Gluten Sensitivity

Eating gluten when you have a sensitivity to gluten can cause issues with the thyroid. The body can create anti-bodies that fight against the thyroid in response to gluten.

HOW CAN WE SUPPORT THYROID

Limit Toxic Exposure

Medication alone will not heal the thyroid. Lifestyle shifts can improve the function of the thyroid and eliminate the symptoms.

#1

Limit Exposure to Environmental Toxins

- Eat organic produce and organic animal products.
- Use only toxin free personal care products (i.e. lotions, makeup, soaps). Check out the Environmental Working Group approved product list.
- Use toxin free household products (i.e. laundry detergent, floor cleaners, dish soap). Check out the Environmental Working Group approved product list.
- Avoid furniture and clothing that has flame resistant material. If you have some furniture with this, you can limit exposure with an organic cover (for example a mattress cover). Fabric covered furniture likely has flame retardant if it was made before 2013. You can look for a tag that says "furniture complies with an **updated** standard, TB117-**2013**", with a checkbox that says no flame retardants to be sure the furniture does not have flame retardant.
- Keep your house and workplace dry, possibly with a dehumidifier. It may be worth it to have an air quality test to assess for mold exposure.
- Use a HEPA air purifier in the rooms you spend the most time in. AirDoctor is a quality brand.
- Bring in air purifying plants (see next page for examples)
- Avoid using plastic containers to store warm food or to reheat food. Also avoid water that is stored in plastic containers. Instead drink filtered water from the refrigerator, sink, or other filtration system.
- If food is packaged in plastic look for the number 1, 2, 4 and 5 near the recycling symbol. These are considered the safest. Avoid plastics with the recycling numbers 3, 6 or 7.

Avoid these chemicals that may be found in your self-care or household products

- Parabens (examples: methylparaben, ethylparaben, propylparaben, and butylparaben)
- Phthalates (if "fragrance" is listed on ingredients, the product likely has phthalates)
- BPA (Bisphenol A)
- Benzene
- PFA (POA, PFHxS/perfluorohexane sulfonic acid, and PFNA/perfluorononanoic.

AIR PURIFYING PLANTS



Snake Plant

Light: Medium and indirect
Water: Less than once per week
Not safe for pets



Spider Plant

Light: Medium and indirect
Water: Moderate moisture
Safe for pets and children



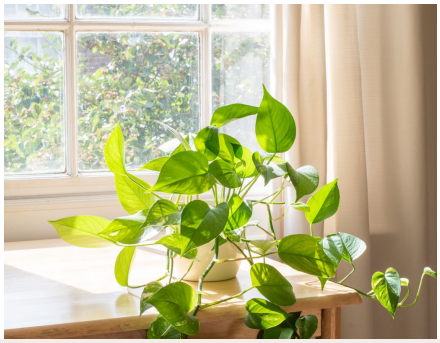
Rubber Plant

Light: Partial shade
Water: Moist, well drained
Not safe for pets and children



Dracaenas

Light: Med/bright and indirect
Water: Moist, well drained
Not safe for pets



Golden Pothos (Devil's Ivy)

Light: Low-Bright and indirect
Water: Less than once per week
Not safe for pets



Bamboo Palms

Light: Bright and indirect
Water: Moderate moisture
Safe for pets



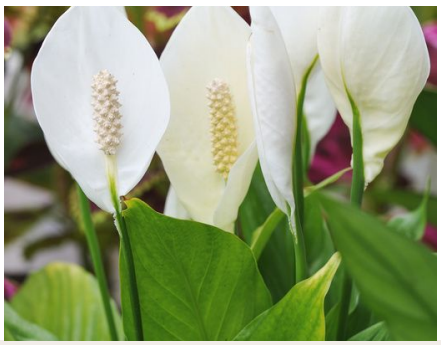
English Ivy

Light: Bright
Water: Minimal moisture
Not safe for pets and children



Chinese Evergreen

Light: Medium/low and indirect
Water: Moderarte moisture
Not safe for pets



Peace Lilies

Light: Medium/low and indirect
Water: Minimal moisture
Not safe for pets and children

HOW CAN WE SUPPORT THYROID

Adjust Your Diet

#2

Get Enough of These Nutrients

It is crucial that you get enough clean protein (from organic plants, organic animal products, and wild caught fish), healthy fats (such as from nuts, seeds, and fatty fish), and fiber from fruits and veggies. See below for top thyroid supporting nutrients and the foods we can get these from.

- Vitamin A: Leafy green vegetables, carrots, sweet potatoes, squash, pumpkin, tomatoes.
- Iodine: Seaweed, shrimp, cod, eggs, prunes.
- Selenium: Brazil nuts, tuna (be sure it is low mercury such as Safe Catch or Wild Plant brands), sardines, eggs, legumes.
- Vitamin D3: Best to get this from natural sun exposure, though supplementation is often necessary. Here is a clean brand that is combined with Vitamin K for optimal absorption and utilization.
- Omega 3's: Salmon, sardines, flaxseed, chia seeds, pasture raised eggs. Supplementation may be necessary , here is a clean brand.
- Zinc: Beef, lamb, shellfish, legumes, hemp seeds, sesame seeds, pumpkin seeds, almonds, cashews, dairy, eggs.

Avoid These As They Can Increase Inflammation and Slow the Thyroid

- Highly processed oils (vegetable oils, corn oil, canola oil, palm oil, rapeseed oil, soybean oil, cotton seed oil)
- Gluten
- Any foods with added sugar or artificial sweeteners
- Nuts and seeds that are "roasted" or fried in added oils
- Fried foods
- Foods with added coloring
- Hot dogs or lunch meat (however there are some new brands for organic lunch meat such as Applegate Organics which is far better than conventional lunch meat)
- Fruit juice
- Non organic fruits and vegetables in which you are eating the skin

HOW CAN WE SUPPORT THYROID

Balance Your Life

#3

Stress Management

I know, easier said than done. However, it is SO important that we keep cortisol levels (our stress hormone) under control for optimal thyroid function. Here are some simple ways to reduce high cortisol levels.

- Avoid artificial light from your phone, computer, or tablet within 2-3 hours of going to sleep. Dim the lights in your home.
- Spend 5-10 minutes every morning outside getting exposure to natural light.
- Avoid caffeinated coffee on an empty stomach. Wait until after breakfast if you wish to drink caffeinated coffee.
- Get seven to eight hours of sleep each night.
- Perform breath work two to three times each day. This is the fastest way to decrease cortisol levels and it only takes 5 minutes! Check out this video for how to do this.
- Be sure you are getting enough Omega 3 fatty acids. If you do not eat fish, you may want to consider a supplement.
- Make social time and physical contact a priority. Laughing with friends and giving hugs has a significant impact on cortisol levels.
- Avoid overloading your schedule. It is okay to say no when you need a break.
- Spend time in nature. Evidence shows that outdoor time, especially away from city life, can improve hormonal balance and reduce stress. Even taking one 10 minute walk outside each day can help reduce elevated cortisol.

